Multimedia Signal Processing - 2nd Module

February, 14 2017

1. **Question (8 pts)**

   Consider two discrete memoryless sources X and Y that emit symbols from the alphabets $\{x_1, x_2, x_3, x_4\}$ and $\{y_1, y_2\}$, respectively. The joint p.m.f. is

	y_1	y_2
x_1	0.15	0.05
x_2	0.05	0.35
x_3	0.1	0.2
x_4	0.05	0.05

   (a) Compute the probability mass functions of symbols of X and Y, separately.

   (b) Design an optimal code to encode single symbols of Y. Justify your answer.

   (c) Design the Huffman code to encode single symbols of X.

   (d) Compute: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (e) Design the Huffman code that jointly encodes one symbol of X and one symbol of Y.

   (f) Compute again: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (g) Compare and comment the results obtained in point (c) and point (e).

   (h) Compute the mutual information $I(X, Y)$.

2. **Question (5 pts)**

   (a) Illustrate the block diagram of a hybrid DCT/DPCM video encoder, commenting each block.

   (b) Explain how motion vectors can be computed, and why they are so important in video coding.

3. **Question (5 pts)**

   (a) Describe the source-system model used to synthesize speech signals.

   (b) Explain how a linear predictive vocoder exploits this scheme, and which are the main parameters sent from the encoder to the decoder.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider the quantization process of a source X emitting symbols from an infinite alphabet with probability distribution function $p(x)$.

   - [T] [F] A scalar uniform quantizer can only be used if $p(x)$ is a uniform distribution.
   - [T] [F] Lloyd-max quantizer always provides better results then a scalar uniform quantizer in terms of distortion.
   - [T] [F] Compander-expander technique does not provide significant benefits if $p(x)$ is a uniform distribution and we plan to use a uniform scalar quantizer.
   - [T] [F] In a Lloyd-Max iteration, once reproduction levels have been updated, decision regions update does not depend on $p(x)$.

   (b) Consider a perceptual audio coding scheme.

   - [T] [F] Typically, frequencies around 4kHz can be quantized with less bits as the absolute threshold of hearing reaches its minimum around those frequencies.
   - [T] [F] Noise components can only mask tones.
   - [T] [F] Considering a masking tone, signal-to-mask ratio depends on the considered frequency.

   (c) Consider a discrete memoryless sources X emitting a sequence of samples x from a finite alphabet.

   - [T] [F] The maximum distortion achievable quantizing symbols from X is given by the source variance σ_X^2.
   - [T] [F] The amount of information in bits of a symbol x is defined as $\log_{10}(\frac{1}{p(x)})$, where $p(x)$ is the probability of observing x.
   - [T] [F] The entropy $H(X)$ is surely greater than zero if we observe two different symbols within the sequence (e.g., $x = 1$ and $x = 0$).

5. **Question (Matlab - 10pts)**

   A grayscale image `I` is encoded using DCT on 8×8 blocks. These DCT coefficients are stored into the matrix `I_dct`.

   (a) Load the variable `I_dct` from `lena.mat`. Display it as a figure. What are the spikes you can observe?

   (b) Decode the image using block-wise inverse 8×8 DCT. Display the resulting image `I_hat`.

   (c) Encode `I_hat` and draw the rate-distortion curves considering two different coding schemes:

   - PCM.
   - Block-wise transform coding based on 8×8 DCT.

   For both schemes consider:

   - scalar uniform quantization with $Q \in \{8, 12, 14, 16, 20\}$ (hint: use `round(.)` and ensure a reproduction level equal to zero).
   - PSNR as distortion measure.
   - Entropy as rate measure.

   (d) Comment the rate-distortion curve obtained (hint: what happens for $Q = 14$?).